Zero — Visual summary of findings

9 November 2026

How students handle multiplication facts with 0 in the A998 fluency test: accuracy, speed, type of error, order of the operands, consistency, and early cognitive predictors.

Maria García · Innovamat Research · source: Zero repository, analysis notebooks and their conclusion cells

4.7 M	170k	×2	78%
responses in the main year (Spain 2025-26)	students, ages 8–15 (replicated in 2024-25)	error rate of 0-facts vs 1-facts, at the same speed	of 0-fact errors answer *n* (0×7 → 7)

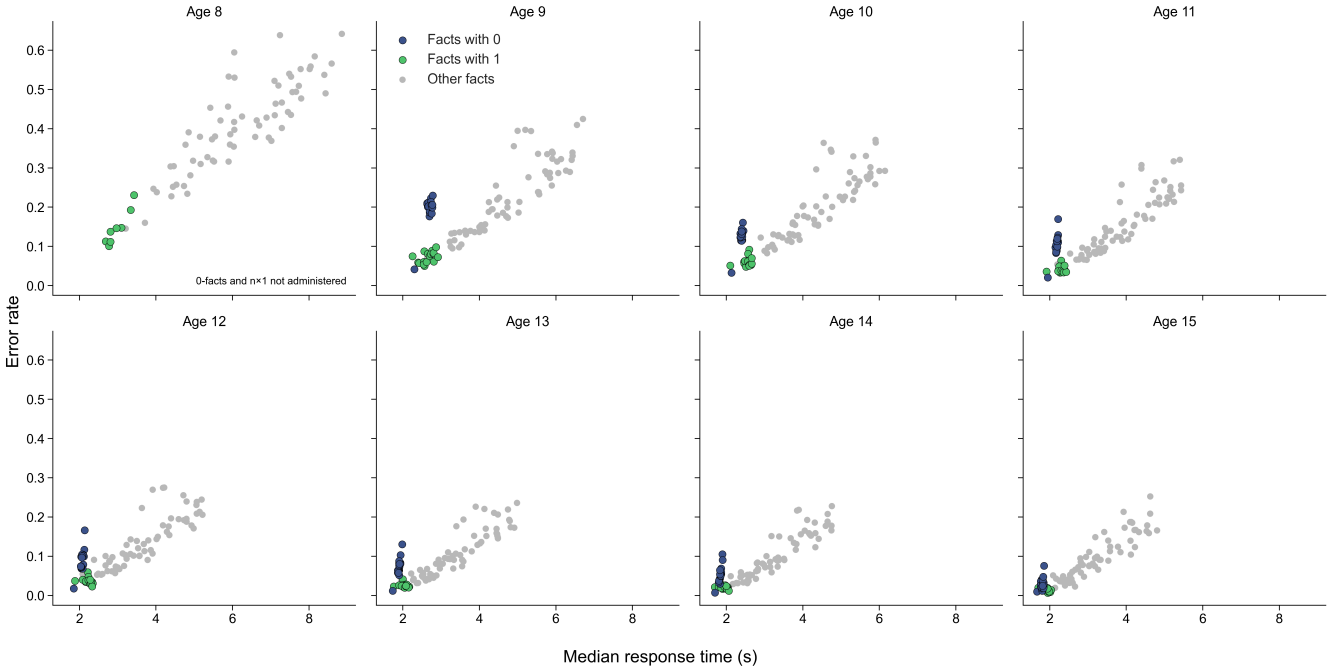
Key findings

- **Fast but wrong.** Facts with 0 are answered as fast as the easiest facts, yet with many more errors than facts of the same speed: speed and accuracy come apart.
- **One dominant mistake.** When wrong, students almost always give the other operand (0×7 = 7): the zero is treated like +0 or like ×1. It is about six times more frequent than the equivalent error on ×1.
- **Order matters.** *n*×0 is slower (+34 ms) and less accurate (−2.4 points) than 0×*n*. The slowdown is shared with the ×1 facts in younger ages, but the accuracy cost is specific to zero.
- **All-or-nothing knowledge.** Getting one ×0 problem right strongly predicts the next one (93% vs 52%), across different operands, orders and even months apart — far more than for ordinary facts.
- **Zero is special, not just "a rule".** The ×1 rule becomes automatic with age; the zero rule stays a conceptual item that students either have or lack.
- **Early predictors (tentative).** Working memory and reasoning at age 7 predict zero-rule use at 9 (≈ 0.13–0.18 SD, *medium* on Kraft's education benchmarks), but not ×1 use. Not yet confirmed.

How to read the numbers. Almost every comparison is made **within the same student** (each student is their own control). Response times use medians because a few very slow answers would distort averages. *p*-values are corrected for multiple comparisons (Holm) when several ages or pairs are tested at once. Main year: Spain 2025-26 (ES_2025); the key results are replicated in Spain 2024-25 (ES_2024). 0-facts are not administered at age 8, and at age 9 only at the end of the year.

1. Facts with 0 are fast but error-prone

How does each fact sit on speed versus accuracy? Each dot is one fact (e.g. 3×7) at one age.



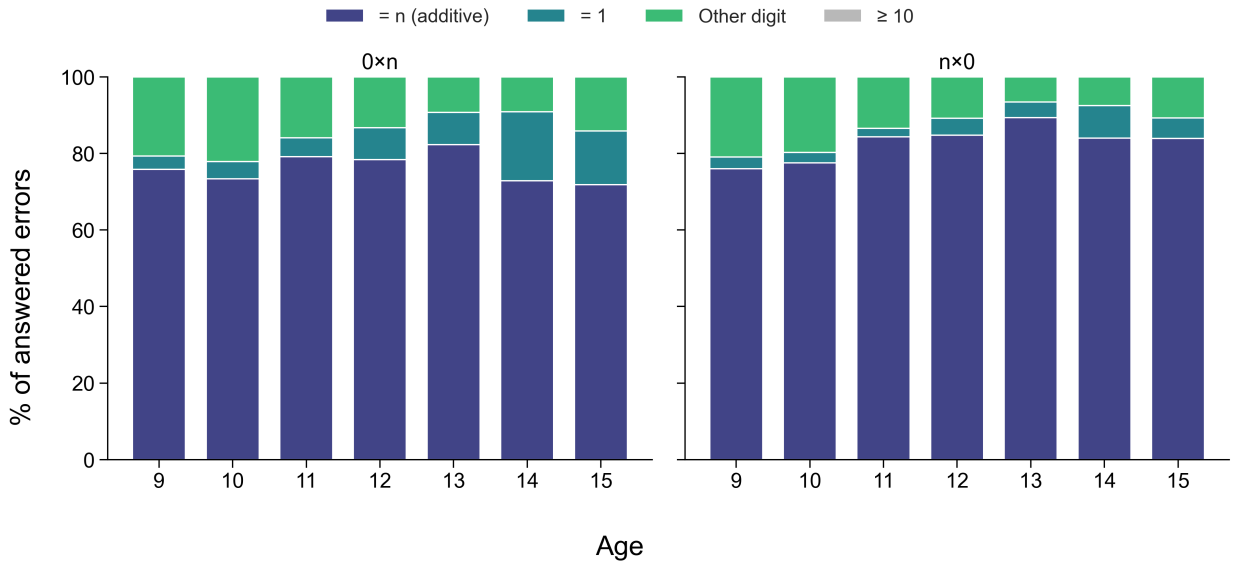
Median response time (correct answers) vs error rate per fact, ES_2025. Grey: facts 2–9; green: facts with 1; navy: facts with 0. The navy column sits at the fast end but well above the grey trend. The one low navy dot is 0×0.

- For ordinary facts, slower means more errors (correlation across facts ≈ .9 at every age).
- Facts with 0 break the pattern: at age 9 a median **20.5% errors vs ≈ 8%** for other facts answered just as fast; still ≈ 6–8 points above the trend at ages 10–13 and ≈ 2 at 15.
- They are as fast as the facts with 1 (median 2.06 s vs 2.22 s, ages 9–15) but have **twice the errors**: 10.4% vs 4.7% (2024-25: 14.6% vs 7.3%). Different at every age (Mann-Whitney over facts, Holm-corrected, $p \leq .001$), and in every collection wave.
- Same picture in 2024-25 and in every country group (Mexico, Italy, Ecuador, Colombia, Chile + Brazil; smaller in the US).

What it suggests. Students answer 0-facts with a quick rule, right or wrong, rather than by slow recall. Being a rule is not enough to explain the errors: the ×1 rule is just as fast and half as error-prone.

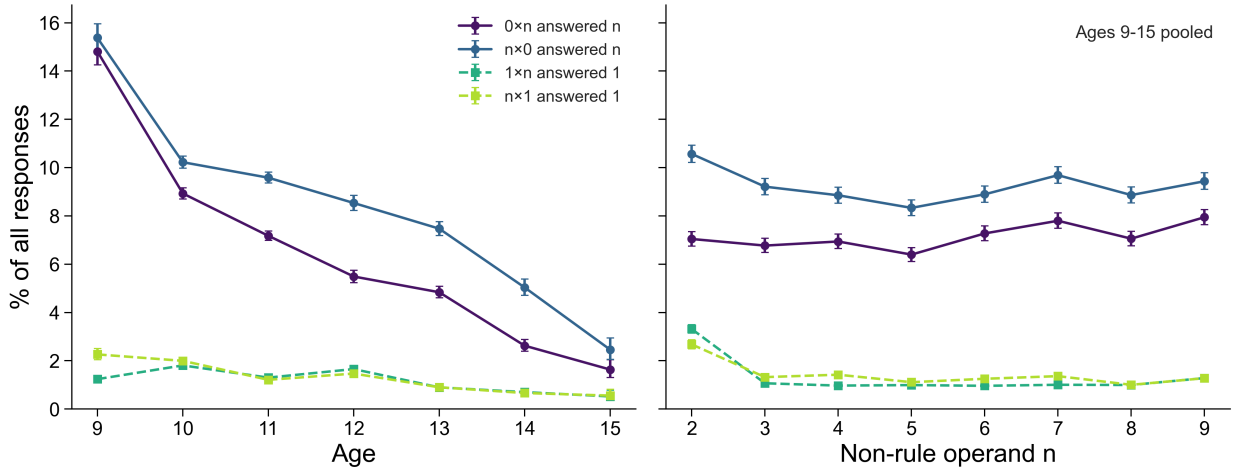
2. What students answer when they get it wrong

Wrong answers on 0×n and n×0, classified by type (ES_2025, ages 9–15, 501,851 responses).



Share of the answered errors by type. Navy: the non-zero operand (0×7 → 7); teal: answering 1; green: another digit. Product-sized answers (≥ 10) never occur.

- Answering the other operand is **72–89% of errors** at every age and in both orders (43,070 responses). Answering 1 is second and grows with age (0×*n*: 3.5% at 9 → 18% at 14).
- Error rate on 0-facts falls from 20.4% at age 9 to 3.1% at 15 — but a minority still makes the mistake in secondary school. 0×0 is almost always right (1–4% errors).



Share of all responses that copy the other operand: n on 0-facts (solid) vs 1 on 1-facts (dashed). Left by age, right by the non-rule operand.

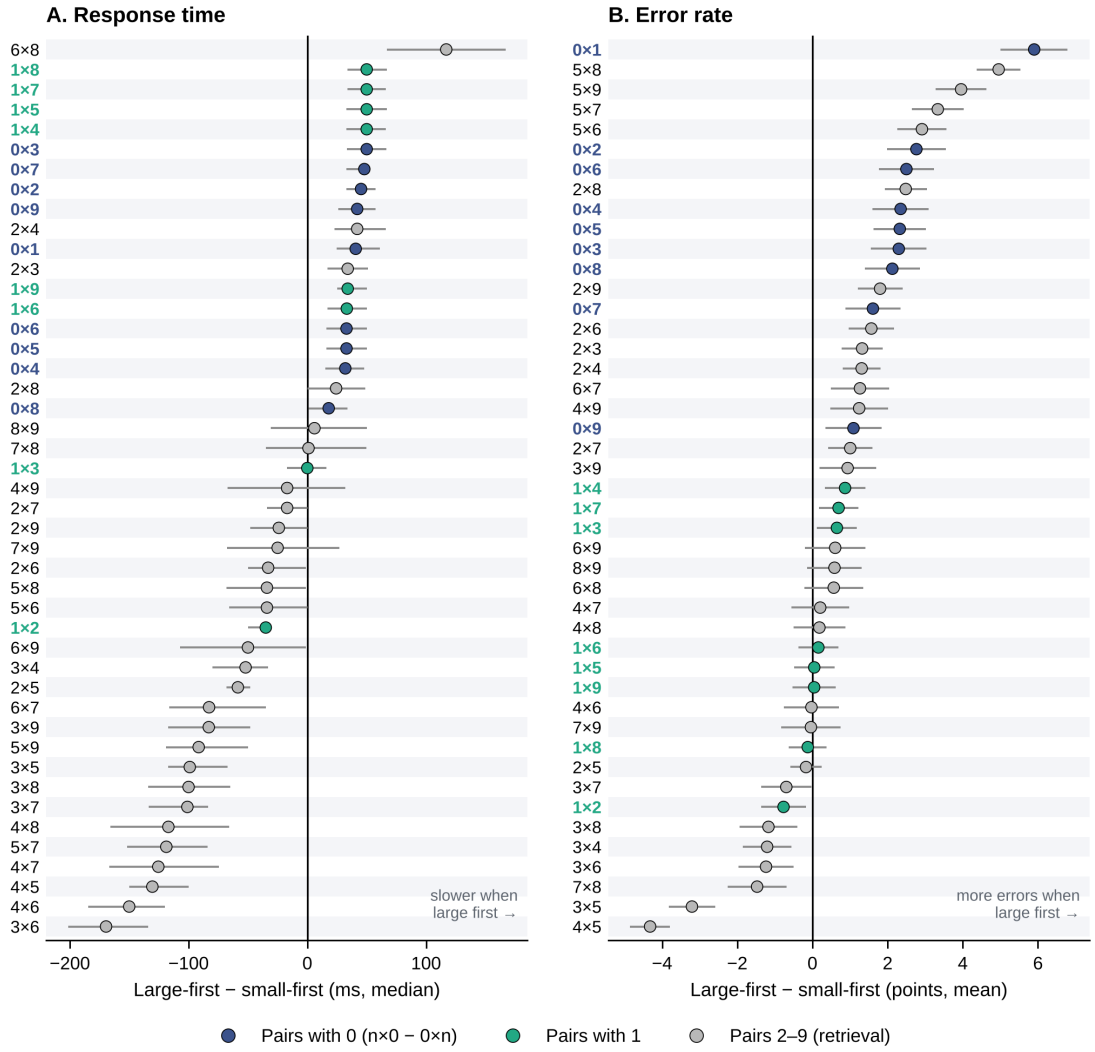
- On 1-facts the equivalent error (1×7 → 1) is rare: **8.2% vs 1.4% of responses** (≈ 6×), 78% vs 29% of errors. On 1-facts mistakes are spread (adding, 7 + 1 = 8, is 17%; applying the zero rule, 6%).
- *n*×0 is worse than 0×*n*: 9.8% vs 7.4% of all responses are *n*; the gap opens after age 9. The worst single fact is 1×0 (14% answer 1).

What it suggests. The zero error is a systematic rule confusion — “multiplying by zero leaves the number unchanged” — not noise or a recall failure. It is a conceptual target for teaching, and more likely when the zero comes second.

3. Order matters: n×0 is slower than 0×n

Correct answers only. Each student answers both orders, so the difference is taken inside each student.

- **+34 ms** median slowdown for *n*×0 [95% CI 33, 42], 70,735 student × age pairs, 53% of students slower (Wilcoxon signed-rank $p < .001$). About 2% of a typical 2.2 s answer.
- Significant at **every age 9–15** (+25 to +59 ms). Six estimators with different assumptions (trial-level regression with student and fact fixed effects, mixed model, one trial per student…) all give +32 to +40 ms; not moderated by proficiency or general speed.
- *n*×0 is also **less accurate**: 89.9% vs 92.2% correct (−2.4 points, odds ratio 0.73), so it is not a speed–accuracy trade-off.



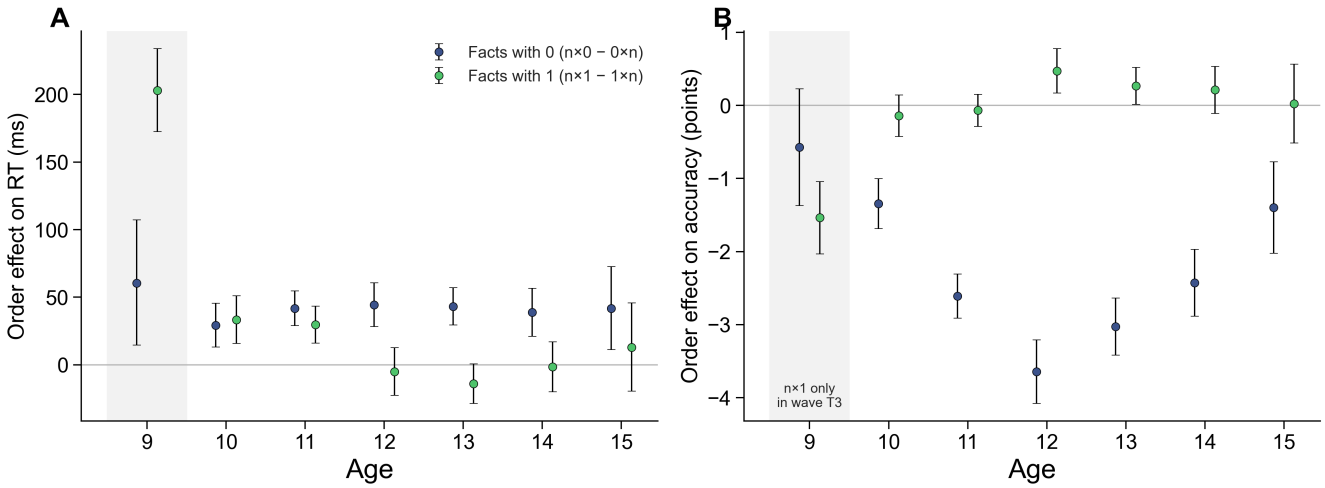
All 45 pairs of different operands (ES_2025), large-first minus small-first, within student, with 95% intervals. A: response time (median, correct answers; 5.8k–22.7k students per pair). B: error rate (mean; 6.8k–30.1k students per pair). For pairs with 0, large-first means n×0.

- **In speed** the cost is a rule effect: large-first is slower in 9/9 zero pairs and 7/8 one pairs, but only 9/28 ordinary pairs (zero = one, $p = .32$; rules vs ordinary, $p < .001$; not explained by difficulty). Replicated in 2024-25 ($\rho = .69$ between years, same sign in 17/17 rule pairs).
- **In accuracy** it is zero-specific: all nine zero pairs have more errors with the zero second (+1 to +6 points); one pairs stay near zero.

What it suggests. The order cost in speed is a general signature of using a rule, and tiny next to the rule's benefit (the rule saves ≈ 1.5 s per answer; the order cost is ≈ 2% of that). The accuracy cost points to something specific to zero: the *n*×0 format invites the “answer *n*” error.

4. Is it zero, or any rule? The ×1 control

Every main analysis repeated with the 1-facts, also solved by a rule (n×1 = n). A pattern present for 0 and absent for 1 cannot be explained by rule use alone.



Order effect (rule operand second minus first) per age, trial-level regression with student and pair fixed effects. A: response time. B: accuracy. Age 9 shaded: n×1 and 0-facts appear only in the last wave, so that age is confounded with time of year.

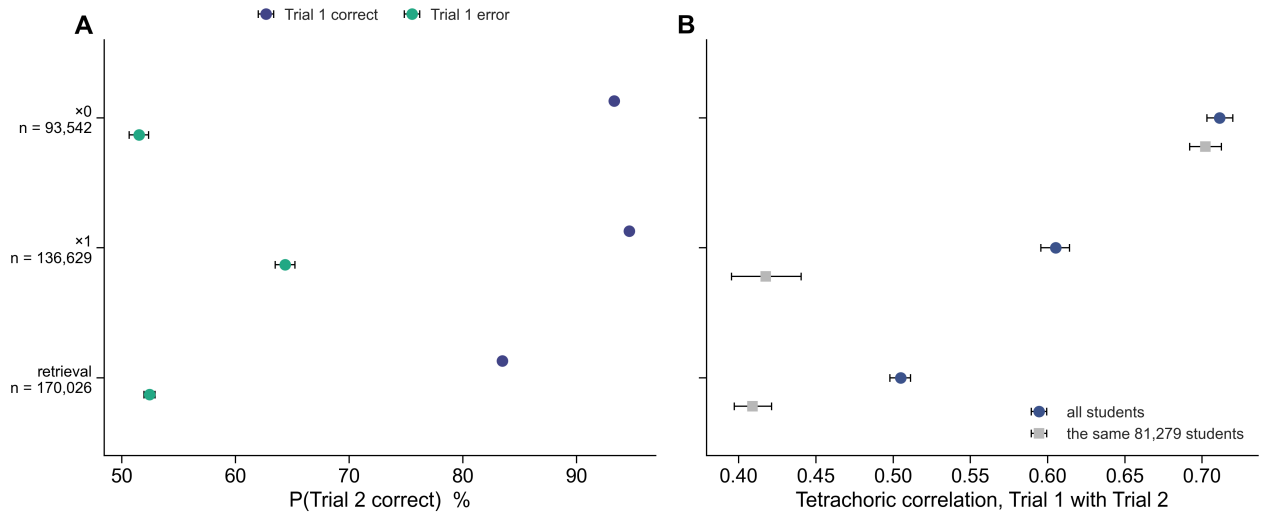
Pattern	Facts with 0	Facts with 1	Specific to zero?
Speed (median RT, ages 9–15)	2.06 s	2.22 s (ordinary facts 3.76 s)	No — both rule-fast
Error rate (ages 9–15)	10.4%	4.7%	**Yes**, ×2 at every age and wave
Copying the other operand	8.2% of responses	1.4% of responses	**Yes**, ≈ 6×
Order cost in speed (ages 10–15)	+1.7%, stable with age	≈ +0.5%, gone from age 12	Partly — only zero after 11
Order cost in accuracy (ages 10–15)	−2.3 points	≈ −0.2 points	**Yes** (replicates 2024-25)
Consistency trial 1 → trial 2	r = .71, stable with age	r = .61, falls to .29–.45	**Yes**, more so with age

- Order-cost tests: fixed-effects regression on 818k trials (speed) and 884k trials (accuracy); the zero vs one difference is estimated inside the same students (both $p < .001$) and replicates in 2024-25. Results are unchanged when “Help” answers are excluded (0.5% of responses).

What it suggests. Rule use explains why 0-facts are fast and part of the early order cost, but not the higher error rate, the operand-copy error, the accuracy cost of *n*×0 or the stable consistency. These are specific to zero, and the 1-facts are the control that rules out generic rule use.

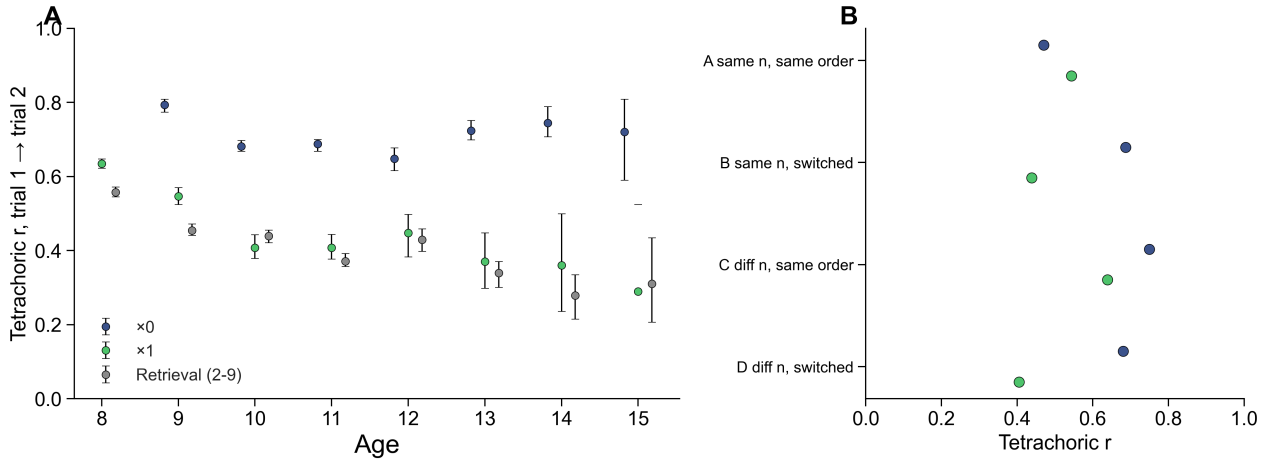
5. Knowing the zero rule is close to all-or-nothing

Within a student, does getting the first ×0 problem right predict the second? (Observational, not causal.)



A: probability of a correct second trial after a correct (navy) or wrong (green) first trial, per problem type. B: tetrachoric correlation between the two trials (comparable across types with different base accuracy); grey squares = the same 81k students in all three types.

- **93.3% vs 51.5%** correct on the second ×0 problem after a correct vs wrong first one (+42 points, odds ratio 13.1; 93,663 students).
- Survives controlling for proficiency, speed, age, response time, operand, order and distance (logistic regression, classroom-clustered: odds ratio 10.7, 94% vs 61%), and inside every proficiency band.
- It is not repetition: it holds when **both** operand and order change (0×7 → 4×0: odds ratio 9.0, 43,511 students); the weakest case is the exact same fact repeated.
- **Specific to zero**: correlation .70 for ×0 vs .42 for ×1 and .41 for ordinary facts on the same students. It still shows across tests a median 102 days apart (+22 points), and A998 gives no feedback after errors.

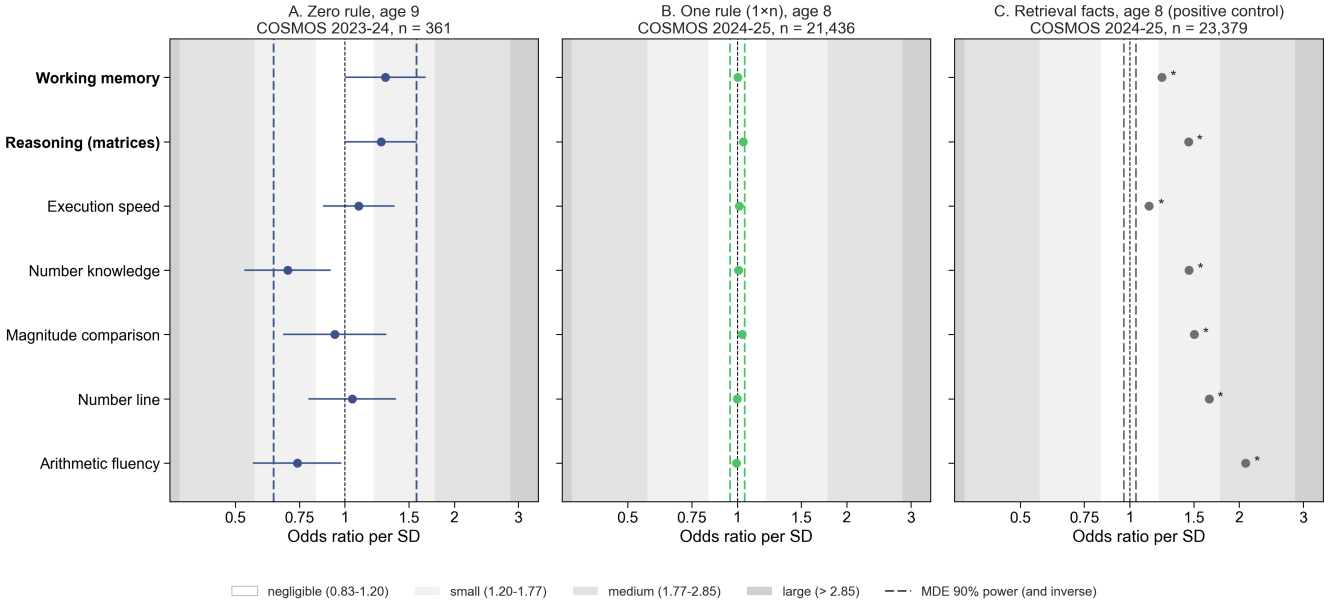


A: trial 1 → trial 2 correlation per age for ×0 (navy), ×1 (green) and ordinary facts (grey). B: the four transfer conditions, ×0 vs ×1.

What it suggests. Students either hold the zero rule or they don't, and they apply it the same way to any operand and in any order. With ×1 and ordinary facts trials become less coupled with age, while ×0 stays tightly coupled — what a rule looks like next to item-by-item recall.

6. Early cognitive predictors (longitudinal, tentative)

Do COSMOS cognitive tasks at ages 6–7 (2023-24) predict zero-rule use in A998 one or two years later?



Odds ratio per SD of each COSMOS factor (combined PRE + POST score). A: zero rule at age 9 (n = 361). B: one rule at age 8 (2024-25 cohort, n = 21,436). C: ordinary facts (positive control). Dashed lines: smallest effect detectable with 90% power. Background bands are the plot's generic OR bands.

- **Working memory and reasoning** predict the zero rule: odds ratio 1.26–1.38 per SD, ≈ **0.13–0.18 SD** on a latent scale — *medium* on Kraft's (2020) education benchmarks (0.05 to < 0.20). Only reasoning PRE survives Holm correction (1.35 [1.08, 1.68], p = .035).
- **Specific to the zero rule**: in the large 2024-25 cohort (≈ 21,800 children) every factor gives an odds ratio of 0.99–1.04 for the one rule — a precise null — while all factors predict ordinary facts strongly.
- **Robust but small sample**: same direction with other outcomes and estimators; selection adjustment (inverse-probability weights) lowers the effects 4–7% and makes intervals include 1. Only 365 children have a 0-fact at age 9, so only effects ≥ 1.55 per SD are reliably detectable.

What it suggests. A modest, zero-specific link with working memory and reasoning that is consistent across methods but not yet confirmed. Planned replication: COSMOS 2024-25 (≈ 10k children with a 0-fact) in the T3 2026-27 wave, with the specification fixed in advance.

Limits to keep in mind

- Observational data: the consistency result cannot separate stable student differences from carry-over between trials.
- Order results come from Spain; item position inside the test session is not modelled, so a warm-up effect could contribute to the order cost.
- At age 9 the 0-facts (and n×1) are tested only at the end of the year; age-9 contrasts are confounded with time.
- About 2–3 correct trials per student and order: effects are population-level shifts within students, not individual diagnoses. 41% of students (short tests) lack a proficiency control in the adjusted consistency models.